Part 7: The Dual Parton Model, Reggeons, and the Pomeron

The Framework for Soft and Diffractive Scattering

PHYS 7502 – Final Reading Assignment

1 What This Part Is About

So far we've focused on *hard* QCD processes: events with a large momentum transfer Q where perturbation theory works. But a huge fraction of what happens at any hadron collider is *soft*: small momentum transfers, no clear partonic subprocess, just two hadrons glancing off each other and (often) staying intact.

The Soft Regime is Huge

At the LHC, the total cross section for pp scattering is about 100 mb at $\sqrt{s} = 13\,\text{TeV}$. The cross section for producing a Higgs boson is about 50 pb. That's a ratio of 2×10^9. The vast majority of pp events are "soft" – elastic scattering, diffractive production, minimum-bias multiplicity. None of this is calculable from first-principles QCD. We need a different framework.

The framework we'll meet here is the **Dual Parton Model (DPM)**, built on the earlier ideas of **Regge theory** and **Pomeron exchange**. This is not a fundamental theory of QCD; it's a *phenomenological* framework that organizes soft scattering data in a remarkably useful way – much like the Eightfold Way organized hadron quantum numbers before the quark model.

2 Back to the Dual Resonance Model

You saw the duality property in Part 6: at high energies, s-channel resonance sums and t-channel exchange sums give the same amplitude. Let's go further now and actually write down the amplitude.

2.1 The Veneziano amplitude

In 1968, Gabriele Veneziano found an elegant closed-form amplitude satisfying duality. For $2 \to 2$ meson scattering:

The Veneziano Amplitude

$$A(s,t) = g^2 \frac{\Gamma(-\alpha(s))\,\Gamma(-\alpha(t))}{\Gamma(-\alpha(s) - \alpha(t))} \tag{1}$$

where Γ is the Euler gamma function, s and t are the usual Mandelstam variables, and $\alpha(x)$ is a linear function

$$\alpha(x) = \alpha_0 + \alpha' x. \tag{2}$$

This is remarkable on many levels. The amplitude is:

- **Explicitly symmetric** in $s \leftrightarrow t$ – duality baked in.
- **Analytic**: for most values of s and t, it's just a well-defined function.
- **Physical**: its poles correspond to actual hadronic resonances.

2.2 Reading the poles

The gamma function $\Gamma(-\alpha)$ has poles at $\alpha = 0, 1, 2, \ldots$ So $\Gamma(-\alpha(s))$ has poles whenever

$$\alpha(s) = J \quad \Leftrightarrow \quad s = M_J^2 = \frac{J - \alpha_0}{\alpha'} \tag{3}$$

For integer $J = 0, 1, 2, \ldots$, this picks out *exactly* the mass-squared values of states on the Regge trajectory (Part 6)!

What the Veneziano Amplitude Tells Us

The Veneziano amplitude *encodes* an infinite tower of particle exchanges, one for each integer spin J on the Regge trajectory. Instead of summing up infinitely many t-channel Feynman diagrams (one per exchanged particle with spin J and mass M_J), you have one compact analytic formula that does the sum for you.

In the modern language, we'd say: the Veneziano amplitude describes the **exchange of an entire Regge trajectory**, or equivalently, a **Reggeon**.

2.3 Reggeon exchange

When you sum all the particle exchanges along a Regge trajectory, the total amplitude at large s behaves as

$$A(s,t) \propto s^{\alpha(t)} \tag{4}$$

This looks like "exchange of a single particle with effective spin $\alpha(t)$". That fictitious particle is called a **Reggeon**. The Reggeon is not a real particle sitting in the Particle Data Group tables – it's a *summed* object that combines contributions from every real particle on the trajectory.

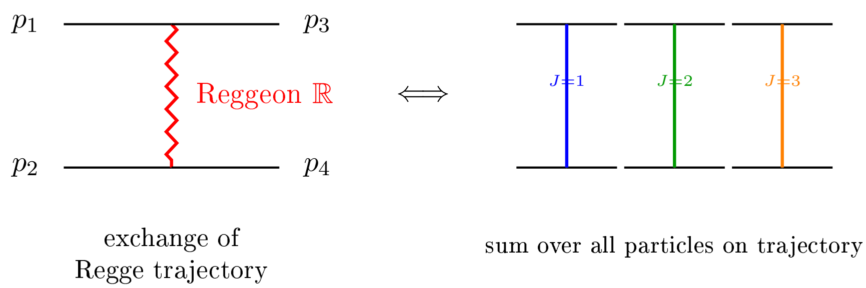


Figure 1: Reggeon exchange. The zigzag on the left represents the sum of infinitely many particle exchanges on the right – all the particles on the Regge trajectory, weighted by their spin. The Reggeon is thus a compact phenomenological object that packages up an infinite series of particle exchanges.

3 The Pomeron: a Special Reggeon

Among the possible Regge trajectories, one is particularly important because it describes the **total cross section at high energy**. Let's work through how this comes about.

3.1 The optical theorem

You may recall the optical theorem from quantum mechanics: the total cross section is related to the *imaginary part* of the forward elastic scattering amplitude:

$$\sigma_{\text{tot}}(s) = \frac{1}{s}\,\mathrm{Im}\,A(s, t = 0) \tag{5}$$

So to find σ_{tot}, we need $\mathrm{Im}\,A(s, 0)$.

3.2 Plugging Reggeon exchange into the optical theorem

Starting from Reggeon exchange, $A(s,t) \propto s^{\alpha(t)}$. Taking the imaginary part at $t = 0$ (some work involving gamma function identities gives a real coefficient times a factor that gives the imaginary part), you find

$$\mathrm{Im}\,A(s, 0) \propto s^{\alpha(0)} \tag{6}$$

Reggeon's Total Cross Section Prediction

Plugging into (5):

$$\boxed{\sigma_{\text{tot}}(s) \propto s^{\alpha(0)-1}} \tag{7}$$

So the exchange of a Regge trajectory with intercept $\alpha(0) = \alpha_0$ gives a total cross section that grows (or falls) with s as a power law.

3.3 What data tells us

Experimentally, hadron total cross sections at high energy have three striking features:

(i) Inelastic processes with **no quantum number flow** dominate the forward region (i.e., particles keep going roughly in the same direction with the same quantum numbers).

(ii) Forward scattering amplitudes are approximately **purely imaginary**.

(iii) Total cross sections approach **nearly constant values** at high energy (actually they slowly rise as $\log^2 s$ or similar, but let's stick with the naive near-constant picture for now).

3.4 What trajectory gives these features?

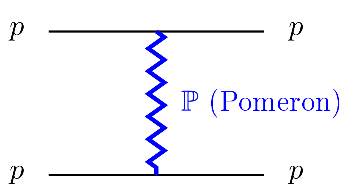
From equation (7), a near-constant σ_{tot} requires $\alpha(0) \approx 1$. But ordinary mesonic trajectories (like the ρ-ω trajectory) have $\alpha(0) \approx 0.5$, giving $\sigma_{\text{tot}} \propto s^{-0.5}$ – a *falling* cross section. That's not what data says.

The Pomeron

To reproduce the observed near-constant total cross section, we postulate a new Regge trajectory with:

- Intercept $\alpha_{\mathbb{P}}(0) \approx 1$ (actually a tiny bit above 1 to account for the slow rise).
- Vacuum quantum numbers (charge 0, strangeness 0, baryon number 0, $I = 0$, $G = +1$, etc.) – so that exchange of this trajectory doesn't change the identity of the particles involved.

This trajectory is called the **Pomeron**, $\mathbb{P}$, named after the Russian physicist Isaak Pomeranchuk.



$pp \to pp$ elastic
or diffractive

Pomeron quantum numbers:

- Intercept $\alpha_{\mathbb{P}}(0) \approx 1.08$
- Vacuum quantum numbers
- Effectively C-even, P-even, $I = 0$
- "Multigluon state" in QCD language

Figure 2: Pomeron exchange in pp elastic (and diffractive) scattering. The Pomeron carries the quantum numbers of the vacuum, so both protons emerge from the exchange with their identities preserved. This is the dominant high-energy scattering process.

3.5 Why vacuum quantum numbers?

Point (i) above – inelastic processes with no quantum-number flow – demands an exchange that doesn't transport charge, flavor, or any conserved quantum number. That's what "vacuum quantum numbers" means. The Pomeron is the carrier of high-energy forward diffraction precisely because it allows the colliding particles to retain their identities while still interacting inelastically.

3.6 What is the Pomeron in QCD?

Pomeron ↔ Multi-Gluon Exchange

In QCD language, the Pomeron is understood as the exchange of a **color-singlet multi-gluon state**. The simplest realization is two gluons in a color singlet (the *BFKL Pomeron*), but more gluons contribute at higher order. Because gluons are flavor-neutral and color-carrying, a color-singlet gluon combination has exactly the quantum numbers we need: vacuum.

So when you see "Pomeron exchange", think: a gluon ladder exchanged between the two protons, color-singlet overall.

4 The Dual Parton Model: Multi-Chain Picture

Now we're ready for the Dual Parton Model itself. The DPM is a *phenomenological* extension of the Reggeon/Pomeron picture into *multi-particle* production.

4.1 Core picture: cut Pomerons and chains

In DPM, a single hadron-hadron collision is modeled as the **cutting of Pomerons**, producing multiple "chains" of particles:

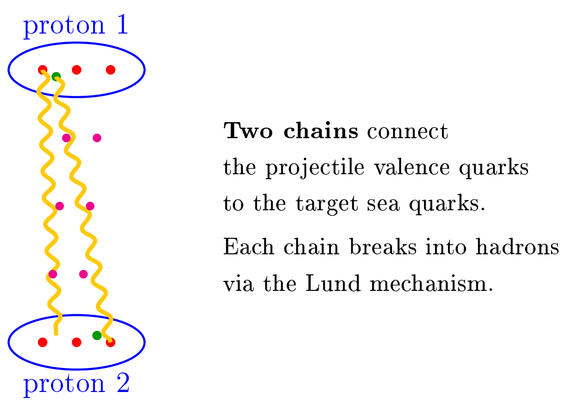


Figure 3: The DPM picture of a *pp* collision. Valence quarks (red) in one proton pair up with sea quarks or diquarks (green) in the other to form strings (chains). Each chain fragments independently into hadrons (purple dots), producing the observed multiplicity. At higher energy, more Pomerons are exchanged and more chains form, naturally producing rising multiplicity.

4.2 Predictions of DPM

The multi-chain picture makes specific predictions:

- **Central plateau.** Each chain produces hadrons with a flat rapidity distribution (Lund!), so the total distribution is flat in the central rapidity region.
- **Rising multiplicity.** Higher energy means more chains, hence more particles. This matches data.

- **Leading hadron spectra.** The valence quark at the top of a chain produces a "leading particle" with a characteristic fragmentation distribution.

4.3 Connection to fragmentation functions

DPM makes predictions for fragmentation functions. For example:

$$xD_{\pi/q}(x) = (1-x) \tag{8}$$
$$xD_{b/q}(x) = x^{\alpha}(1-x)^3 \tag{9}$$

for pions and baryons from a quark. Different variants of DPM (like the VENUS code by Werner) refined these and generalized to diquark fragmentation as well. These predictions are qualitatively consistent with measured FFs, and they provided one of the early cross-checks of the DPM framework.

5 Related Concepts You Should Know About

5.1 Vector Meson Dominance (VMD)

In photon-induced processes (like electroproduction $ep \to ehX$), the photon can fluctuate into a virtual $q\bar{q}$ pair. Since the photon is a vector particle ($J^P = 1^-$), the dominant fluctuation is into light vector mesons with the same quantum numbers:

$$\gamma \rightleftharpoons \rho^0, \omega, \phi, J/\psi, \ldots \tag{10}$$

Vector Meson Dominance, VMD

In the photoproduction of hadrons, the photon effectively "acts like" a superposition of light vector mesons. This is called **Vector Meson Dominance** and explains why photoproduction cross sections are dominated by $\rho, \omega, \phi, J/\psi$ in the final state.

VMD was the first bridge between photon physics and hadron physics. It's roughly the "photon-side" analog of the Pomeron picture for diffractive hadron scattering.

5.2 The Color Dipole Model

A close cousin of VMD. In the Color Dipole Model, the incoming photon fluctuates into a $q\bar{q}$ "color dipole" before interacting with the target:

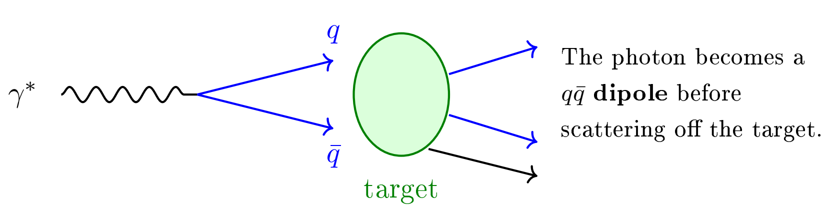


Figure 4: The color dipole picture. In this framework, a photon (left) briefly fluctuates into a $q\bar{q}$ pair – a color dipole – which is what actually scatters off the target. The fluctuation lifetime is controlled by the photon's virtuality Q^2.

Why the Color Dipole Model is Interesting

In the color dipole model, the **dipole cross section** σ_{dip} (which describes how a $q\bar{q}$ dipole scatters off the target) is the *non-perturbative* object. The hadronic structure functions, usually thought of as non-perturbative, become perturbatively calculable in this framework (given the dipole cross section).

This is the *opposite* of the usual factorization story, where PDFs are the non-perturbative input and the hard cross section is perturbative. Whether this is a better framework depends on the kinematic regime – the color dipole approach is especially useful at small x (where the photon fluctuation has time to develop before interacting).

5.3 Color Transparency

An offshoot prediction of the color dipole model: in certain kinematic regimes, a small-sized $q\bar{q}$ dipole passes through a nucleus without interacting significantly, because its color field largely cancels at large distances. This phenomenon is called **color transparency**. It's been looked for experimentally with mixed but generally supportive results.

6 Putting the Soft-Scattering Picture Together

Here's the conceptual hierarchy you've just walked through:

Duality: s- and t-channel sums agree

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Veneziano amplitude encodes this

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Reggeon = whole trajectory exchanged

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Pomeron = Reggeon with vacuum qn, $\alpha_0 = 1$

↓

DPM = cutting Pomerons into chains

↓

Multi-particle production at soft s

Each step builds on the one above. At the bottom of the chain is the thing we actually see at colliders: multi-particle production in soft-scattering events.

7 Summary of Part 7

Main Takeaways

1. The **Veneziano amplitude** $A(s,t) = g^2\Gamma(-\alpha(s))\Gamma(-\alpha(t))/\Gamma(-\alpha(s)-\alpha(t))$ encodes duality explicitly and has poles at the physical Regge-trajectory masses.

2. Summing over all particles on a Regge trajectory gives effective **Reggeon exchange**, with $A \propto s^{\alpha(t)}$ at large s.

3. The **Pomeron** is the Regge trajectory with vacuum quantum numbers and intercept $\alpha_{\mathbb{P}}(0) \approx 1$. Its exchange gives near-constant total cross sections at high energy, matching data.

4. In QCD, the Pomeron corresponds to a **color-singlet multi-gluon exchange**.

5. The **Dual Parton Model** generalizes Pomeron exchange to multi-particle production by cutting Pomerons into chains. Each chain fragments via the Lund mechanism.

6. Related concepts: **Vector Meson Dominance** (photons acting like light vector mesons), the **color dipole model** (photon fluctuating into a $q\bar{q}$ pair before interacting), and **color transparency**.

7. This framework is not derivable from first-principles QCD, but it organizes soft scattering data in a remarkably useful way – much like the Eightfold Way organized hadron quantum numbers before the quark model.